OpenTutor — Novelty & Research Contribution

Project: Personalized Learning AI

OpenTutor is an open-source adaptive learning system that builds an explicit model of a learner's mastery, misconceptions, confidence and forgetting, then uses that state to decide what the learner should study next.

1. Core Novelty

The novelty is **not** simply using an LLM to teach. Modern AI systems can already summarize, answer questions and generate quizzes. OpenTutor treats personalization as a measurable ML problem: **infer learner state** → **identify why the learner is struggling** → **select the next learning activity** → **observe the result** → **update the learner model.**

2. Difference From a Generic AI Tutor

Capability	Generic LLM / AI Tutor	OpenTutor
Answer questions	Yes	Yes
Generate explanations	Yes	Yes
Generate quizzes	Yes	Yes
Persistent learner model	Limited / application-dependent	**Explicit model**
Concept-level mastery	Not the core objective	**Core objective**
Misconception tracking	Not guaranteed	**Core objective**
Prerequisite-aware path	Not guaranteed	**Core objective**
Adaptive difficulty	Possible	**Reproducible policy**
Open training dataset	No	**Yes**
Reproducible experiments	No	**Yes**
Ablation-based evaluation	Usually unavailable	**Yes**

3. Novel Component #1 — Explicit Learner State

For each learner, OpenTutor stores structured state: mastery per concept, confidence, recent mistakes, misconception labels, response history, time since review and prerequisite readiness. Example: Safe State 32% mastery, Banker's Algorithm 42%, Circular Wait 84%.

4. Novel Component #2 — Misconception-Aware Personalization

Instead of only detecting that an answer is wrong, the system attempts to identify *why*. For example, a learner may confuse safe state with no process waiting, or starvation with deadlock. The next activity is selected to repair that misconception.

5. Novel Component #3 — Adaptive Learning Policy

The recommendation engine chooses among actions such as introducing a prerequisite, giving a simpler question, reviewing a weak concept, presenting a contrasting example, or increasing difficulty. This policy becomes an experimental ML component.

6. Novel Component #4 — Knowledge / Prerequisite Graph

The learner model is connected to a concept graph. If a learner struggles with Banker's Algorithm, the system can inspect prerequisites such as safe states, resource allocation and deadlock conditions before recommending advanced material.

7. Novel Component #5 — Forgetting-Aware Revision

Mastery is not treated as permanent. Time since practice and performance history influence when previously learned concepts should be reviewed.

8. Main Research Question

Does an explicit, misconception-aware learner model produce better learning sequences and measurable learning outcomes than static sequencing or generic LLM-generated practice?

9. Experiments That Establish the Novelty

Experiment	Comparison	What it demonstrates
E1	Base LLM vs fine-tuned misconception model	Whether specialization improves misconception detection
E2	Static questions vs adaptive questions	Whether personalization improves sequencing
E3	Difficulty-only vs misconception-aware adaptation	Whether knowing the reason for failure matters
E4	No prerequisite graph vs graph-aware model	Whether curriculum structure improves recommendations
E5	No forgetting model vs review-aware model	Whether time-aware revision improves retention
E6	Ablate each component	Which parts actually create gains

10. What We Should Claim

We should **not** claim that OpenTutor is smarter than ChatGPT or Gemini. The defensible contribution is an open, reproducible framework for **learner-state modeling and adaptive educational sequencing**, with explicit models, datasets and experiments that test whether each personalization component improves learning.

11. Open-Source Contribution

Release the learner-interaction schema, misconception taxonomy, training code, learner-model implementations, recommendation algorithms, evaluation benchmark, experiment configurations and trained model artifacts. Researchers can reproduce results or replace individual components.

12. Final Novelty Statement

OpenTutor is an open-source personalized learning engine that explicitly models what a learner knows, why they make mistakes, how their knowledge changes over time, and what learning activity should come next — with each component evaluated through reproducible ML experiments.

13. Scope Decision

For v1, focus on **Computer Science education**. A focused domain provides cleaner concepts, prerequisite graphs, misconception labels and benchmarks, making the contribution easier to measure and the final system more credible.